

EXAMPLE 5.17

An airline wants to evaluate the depth perception of its pilots over the age of fifty. A random sample of $n = 14$ airline pilots over the age of fifty are asked to judge the distance between two markers placed 20 feet apart at the opposite end of the laboratory. The sample data listed here are the pilots' error (recorded in feet) in judging the distance.

2.7 2.4 1.9 2.6 2.4 1.9 2.3
2.2 2.5 2.3 1.8 2.5 2.0 2.2

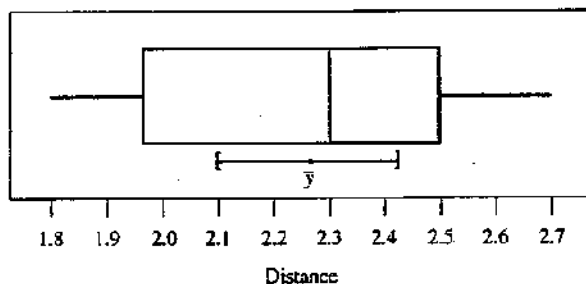
Use the sample data to place a 95% confidence interval on μ , the average error in depth perception for the company's pilots over the age of fifty.

SOLUTION Before setting up a 95% confidence interval on μ , we must first assess the normality assumption by plotting the data in a normal probability plot or a boxplot. Figure 5.20 is a boxplot of the 14 data values. The median line is near the center of the box, the right and left whiskers are approximately the same length, and there are no outliers. The data appear to be a sample from a normal distribution. Thus, it is appropriate to construct the confidence interval based on the t distribution. You can verify that

$$\bar{y} = 2.26 \text{ and } s = .28$$

FIGURE 5.20

Boxplot of distance (with 95% t confidence interval for the mean)



Referring to Table 2 in the Appendix, the t -value corresponding to $\alpha = .025$ and $df = 13$ is 2.160. Hence, the 95% confidence interval for μ is

$$\bar{y} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \text{ or } 2.26 \pm 2.160 \frac{.28}{\sqrt{14}}$$

which is the interval $2.26 \pm .16$, or 2.10 to 2.42. Thus, we are 95% confident that the average error in the pilots' judgment of the distance is between 2.10 and 2.42 feet.